Multi-Physics Simulation (Application in MATLAB & Simulink)

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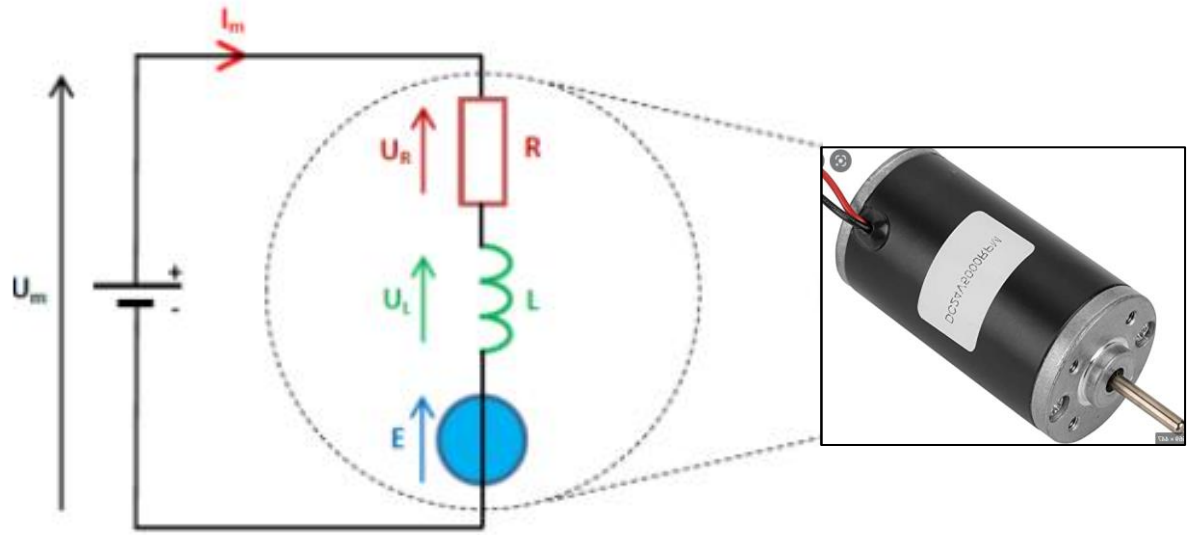
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Getting started: modeling

The system modeling process can be broken down into 6 stages:

- System definition
- Identification of variables/parameters (inputs/outputs)
- Setting up equations
- Block diagram design
- System simulation
- Model validation and/or recalibration == > tests

DC Motor Power Supply Diagram



Objective 1 determine the motor current

$$E = Counter - electromotive\ force = K_e \times \omega$$

K_e *electormotive force constant*

Kt *motor torque constant*

ω *motor rotation speed*

R *motor resistance*

L *inuctance*

i *motor current*

Data

U = 24 V

L = 2^{e-3} H

R = 2 Ohm

K e = 0,8 V/rads/s

DC motor

$u_R = R \cdot i$

$u_L = L \cdot \frac{di}{dt}$

$E = K_e \cdot \omega$ (ω = ang. speed motor)

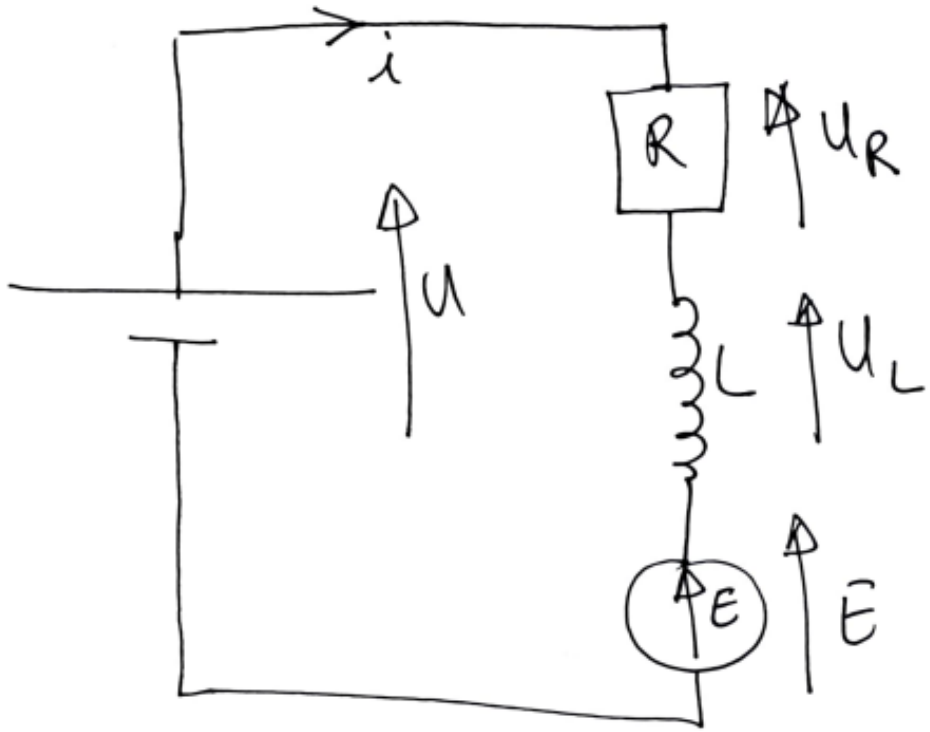
Kirchhoff's Voltage Law :

$\vec{U} = \vec{U}_R + \vec{U}_L + \vec{E}$

$\Rightarrow U = Ri + E + L\frac{di}{dt}$

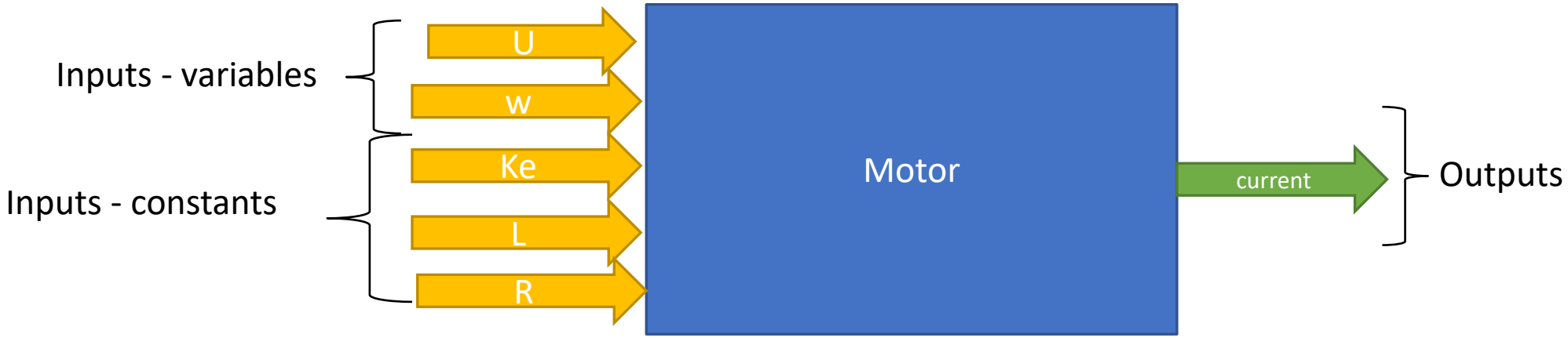
$\Rightarrow \boxed{i = \int \frac{1}{L}\left[U - Ri - E\right]}$ → to Model in Simulink

Electric Diagramm + Power Supply



1) Define the system

2) Variables, inputs, outputs, constants



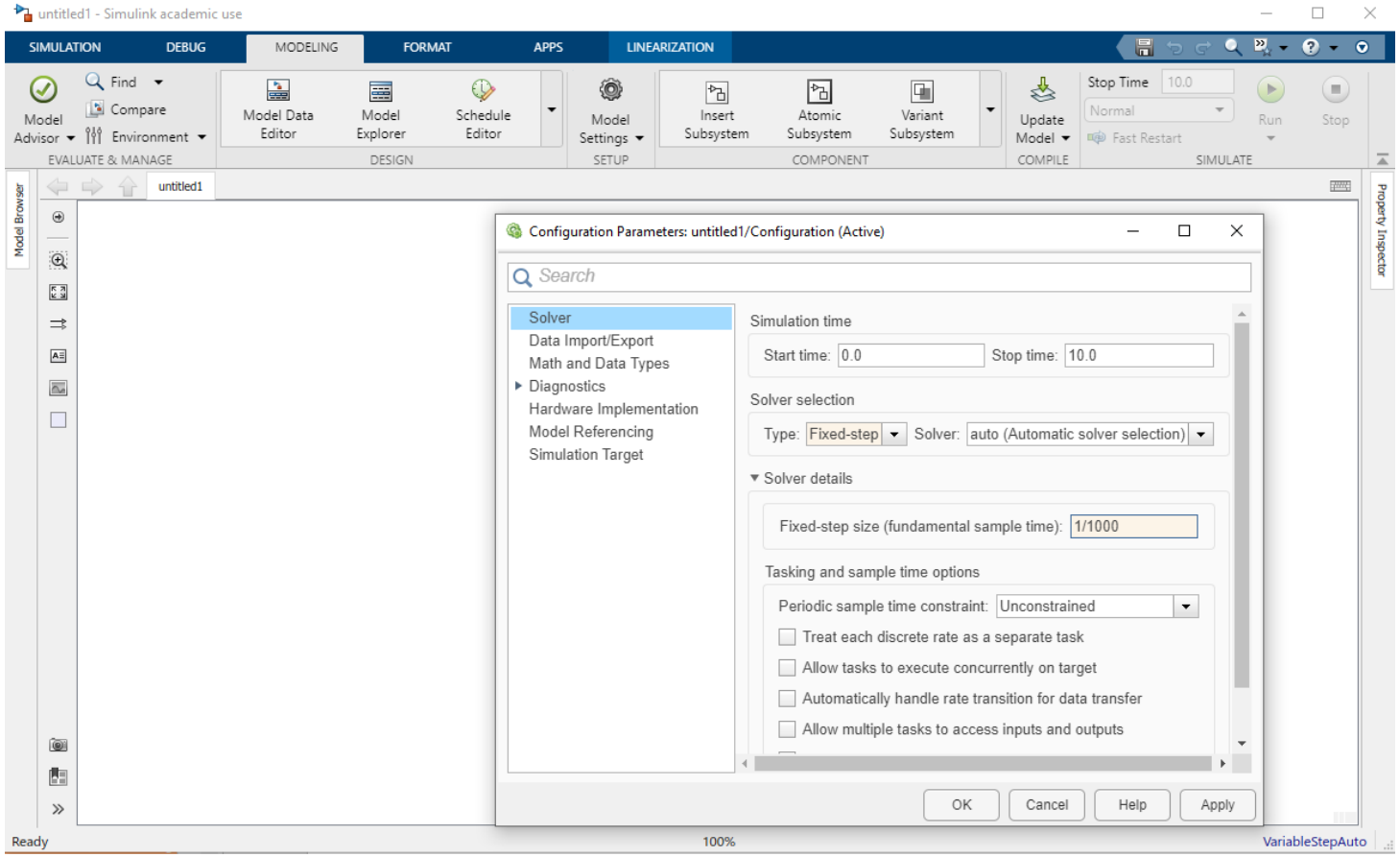
3) Equation

Voltage supply : $U = E + Ri + L\frac{di}{dt} \Rightarrow i = \frac{1}{L}\int(U - Ri - E)dt$

4) Block Diagram + Model Design

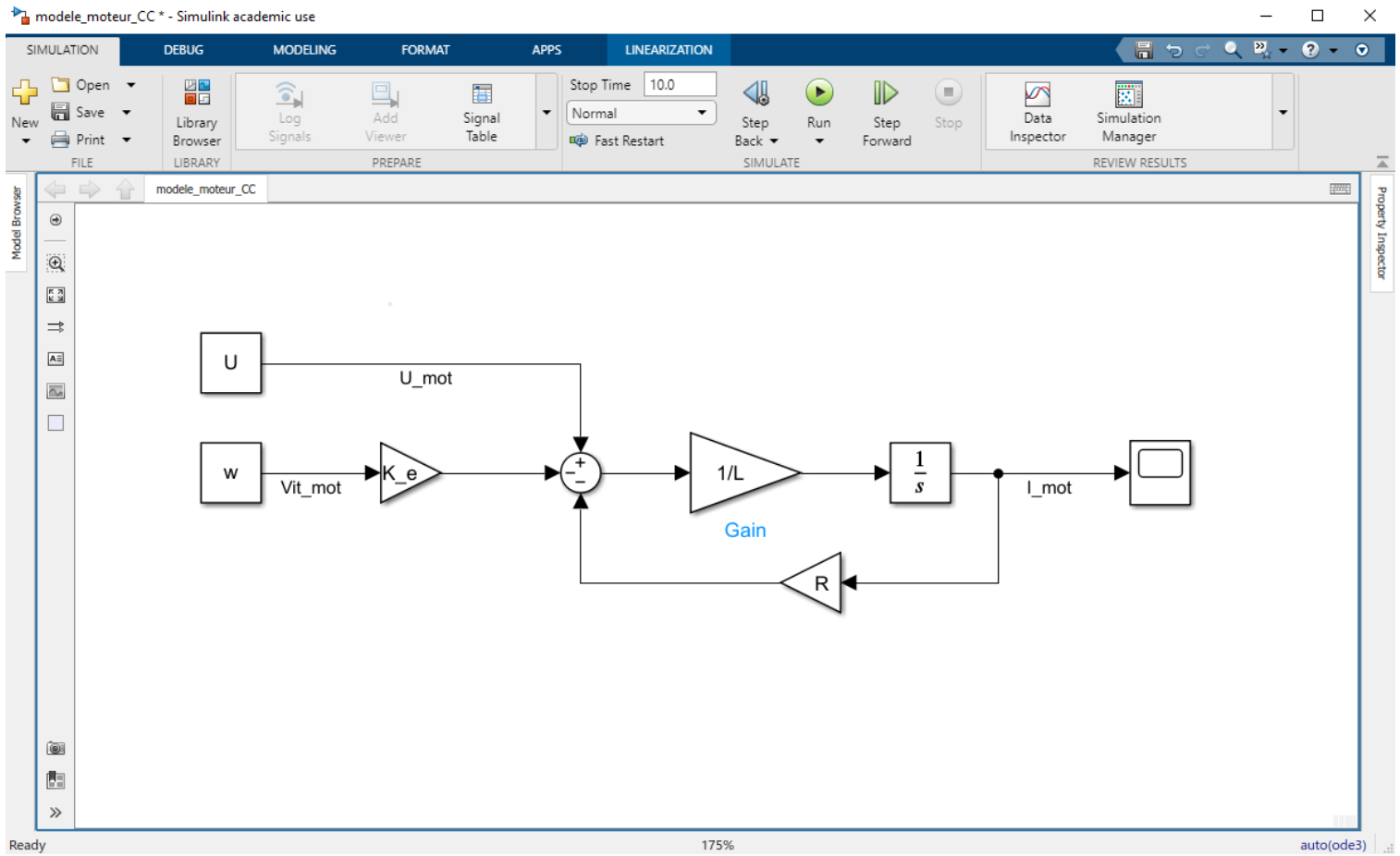
$$i = \frac{1}{L}\int (U - Ri - E)dt$$

Creating a new Simulink model + Setting the solver to fixed-step at 1 kHz



4) Block Diagram + Model Design

$$i = \frac{1}{L}\int (U - Ri - E)dt$$



5) Create an .m file for the parameters + run the file

Editor - C:\Users\Moi\Documents\Pro\Cours_EFREI\ST2SMP_simulation_mutltiphysique\Moteur_Courant_continue\parametres_moteur_CC.m

parametres_moteur_CC.m

```
% date - Auteur -
% ce fichier permet d'intialiser les paramètres du monteur CC et de son
% alimentation

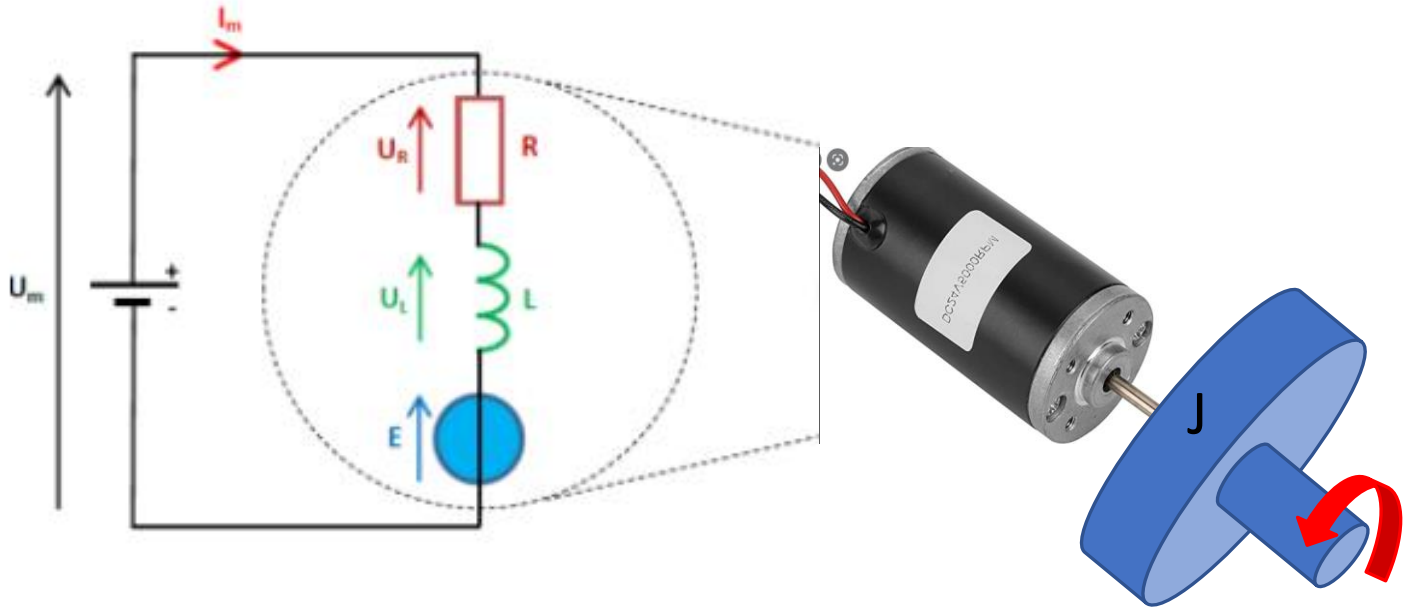
clear all
clc
close all

%% Paramètres electriques
U   = 24    ; % (V) tension au bornes de l'alimentation du moteur
L   = 2e-3  ; % (Henry) inducantance bobinage moteur
R   = 2     ; % (Ohm) Resistance bobinage moteur
K_e = 0.8   ; % (V/rads/s) constante de force contre electromotrice

%% paramètres physiques
w = 3 ; % (rad/s) vitesse moteur
```

6) Simulation + results

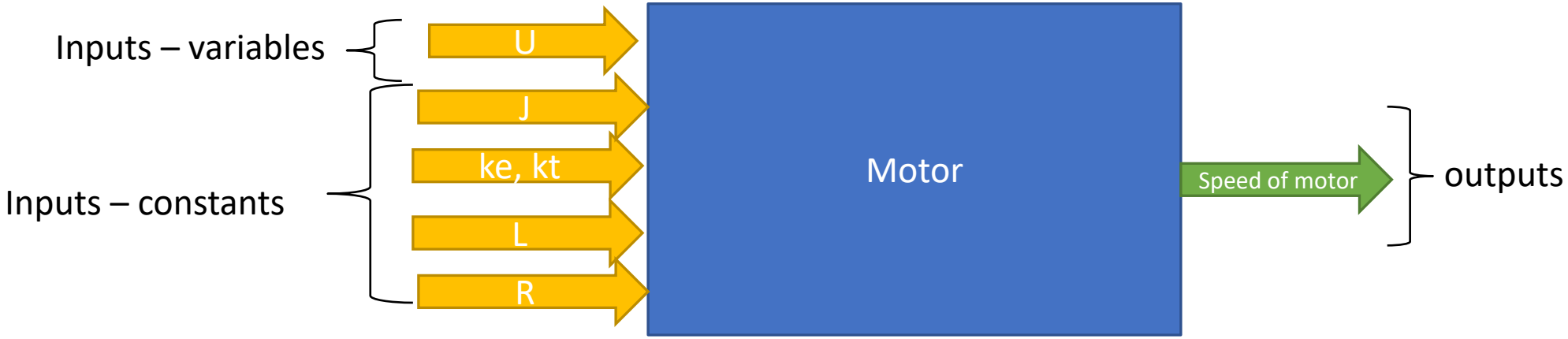
DC Motor + Inertial Load J



Objective 2 determine the motor speed J = 5^e-2 kg.m²

- **Part A: Neglecting friction,**
- **Part B: Taking friction into account**

DC motor


Inputs – variables
U
Inputs – constants
J
ke, kt
L
R
Motor
Speed of motor
outputs

DC motor

Newton's law $\sum \vec{F} = m\vec{\gamma}$, in rotation $\sum \overrightarrow{\boldsymbol{Couple}} = \boldsymbol{J}\overrightarrow{\ddot{\boldsymbol{\theta}}}$

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In the absence of friction, the only applied torque is the motor torque $= Kt \times I$

Newton's law $\sum \vec{F} = m\vec{\gamma}$, in rotation $\sum \vec{Couple} = J\vec{\ddot{\theta}}$

In the absence of friction, the only applied torque is the motor

$$torque = Kt \times I$$

$$\Rightarrow kt \times I = J\,\ddot{\theta} \Rightarrow \theta = \iint \frac{kt}{J} I\,dt$$

Taking Friction into Account

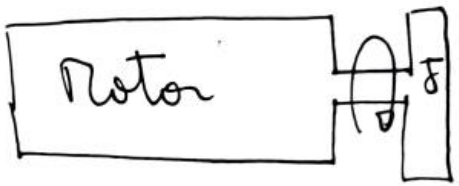
C_sec = -constante1*sign($\dot{\theta}$)

C_visq =-constante2* $\dot{\theta}$

Torque applied by the motor :

$T_{motor} = K_t \times I$ → Rotor Current

↙ Rotor torque Constant



Rotor's speed calculation ?

Newton Law's in rotation : $\sum \vec{T} = J.\ddot{\theta}$ → Angular acceleration

↙ Inertial load

Case ① without friction : $T_{mot} = J\ddot{\theta}$

$K_t.I = J\ddot{\theta} \Rightarrow \boxed{\theta = \iint \frac{K_t.I}{J}}$

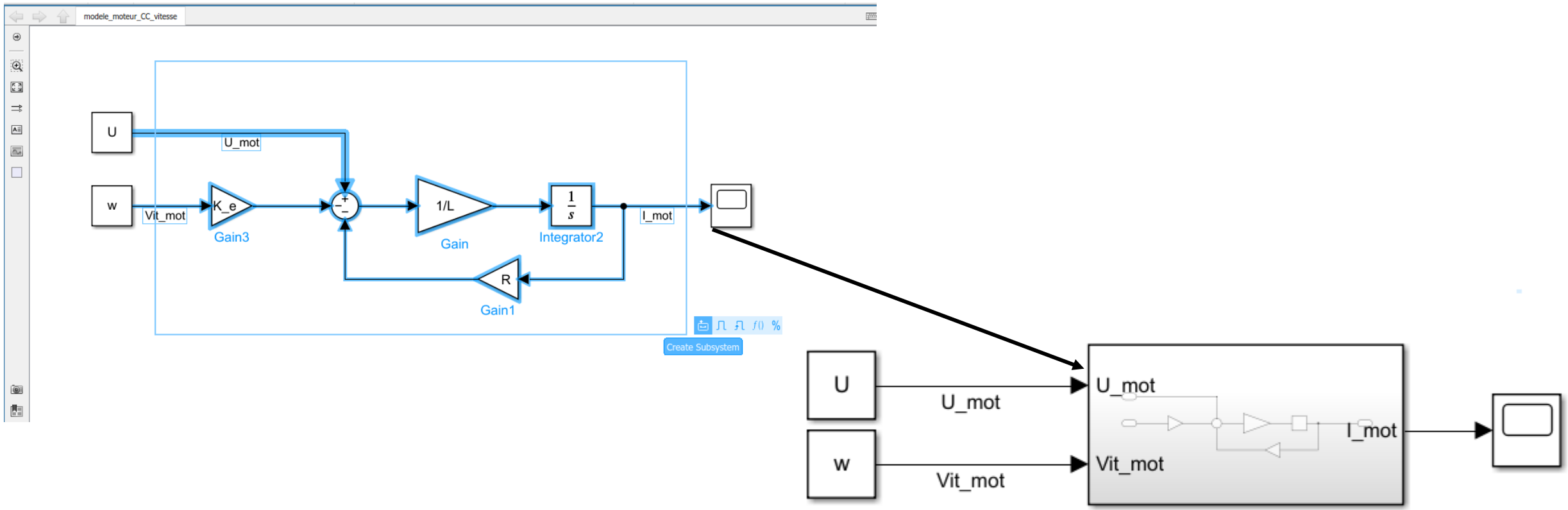
Case ② with friction : $T_{mot} - T_{coulomb} - T_{visq} = J.\ddot{\theta}$

$\Rightarrow K_t.I - C_{sec}.sign(\dot{\theta}) - C_{visq}.\dot{\theta} = J.\ddot{\theta}$

$\Rightarrow \boxed{\theta = \iint \frac{1}{J}\left[K_t.I - C_{sec}.sign(\dot{\theta}) - C_{visq}\dot{\theta}\right]}$

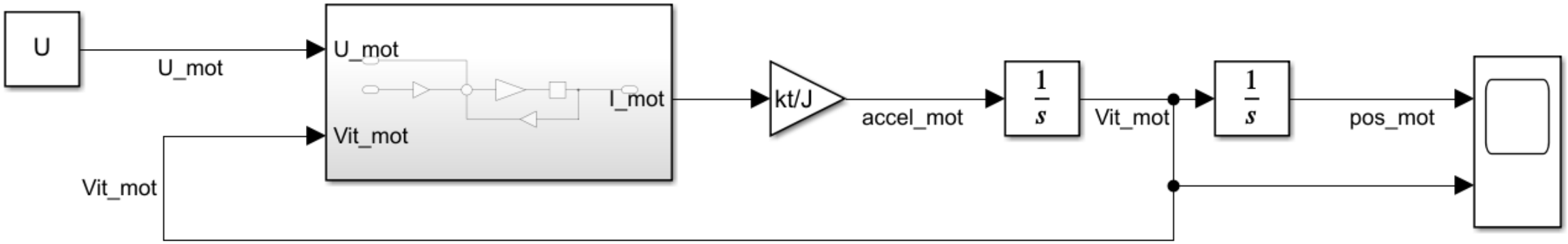
DC motor

Without frictions



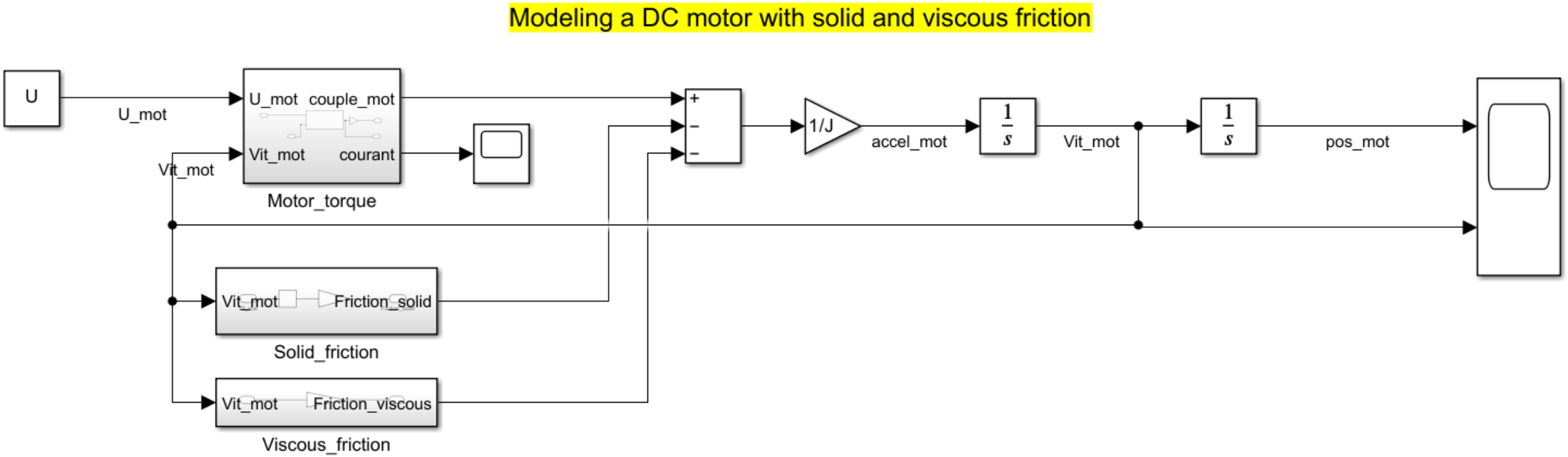
DC motor

Without frictions



DC motor

With frictions



With frictions

